PSG COLLEGE OF TECHNOLOGY, COIMBATORE – 641 004
DEPARTMENT OF COMPUTER APPLICATIONS
AY 2026-2027 MCA First Year Semester 1
26MX17 WEB APPLICATION DEVELOPMENT
Design Abstract

I. Team Detail

Batch No.	Roll No.	Name of the student	Name of the Faculty Guide
26MX	26MX325	KIRUBHAKARAN K	DR. ILAYARAJA N
	26MX337	POOJA M	
	26MX347	SRINITHISH J K	

II. Title of the Application

TrustVault - Smart Digital Identity and Document Verification Portal for college students and faculty.

III. Abstract

TrustVault is a secure web application in which MCA students keep their identity and academic documents (PAN, Aadhaar, driving licence, marksheets and certificates) in one encrypted vault and share them with faculty only with their consent. Unlike general cloud storage such as Google Drive, every file is encrypted with AES-256-GCM, sign-in needs three factors (password, an email code and Google Authenticator) and a separate PIN protects the vault. Each upload is checked automatically (format, Aadhaar checksum, OCR and signs of editing) and is verified only when every check passes. Faculty request documents from one student or a whole batch such as 26MX, each shared document opens only once, and every upload and decision is sealed in a SHA-256 hash-chained ledger. Built with React, Node.js, Express and MongoDB, the application makes document verification faster, safer and transparent for the department.

IV. Existing System (Indian & Global)

- **DigiLocker (India)**

  DigiLocker is the digital document wallet of the Ministry of Electronics and Information Technology, Government of India, under the Digital India programme. Citizens store and share government-issued documents such as Aadhaar, PAN, driving licence and board marksheets, which are fetched directly from the issuing departments.

- **National Academic Depository - NAD (India)**

  The National Academic Depository is an online store of academic awards such as degree certificates and marksheets, lodged by universities and boards and offered through DigiLocker, so that students, employers and institutions can access and verify them online.

- **Google Drive (Global)**

  Google Drive is a cloud storage service in which users keep files of any kind and share them through links or with chosen people. It is widely used by students to send scanned certificates, but it does not check whether an uploaded document is genuine.

- **Parchment (Global)**

  Parchment is a credential platform used by schools and universities, mainly in the United States, to issue and send official digital transcripts, diplomas and certificates that receiving institutions and employers can verify.

V. Proposed System

The proposed system, TrustVault, is a secure, college-exclusive web application in which MCA students store their identity and academic documents and share them with faculty only with consent. Sign-up is limited to valid MX roll numbers for students and to department codes or email invitations for staff, and every sign-in passes three steps: password, an email code and Google Authenticator. Uploaded files are encrypted with AES-256-GCM and checked automatically (PAN format, Aadhaar Verhoeff checksum, driving licence format, OCR and originality traces); a document is verified only when every check passes, otherwise the student sees which check failed. Faculty request documents from one student or a whole batch, the student shares all, some or none, and each shared document opens once within 24 hours. The administrator manages users and reviews the audit log but cannot open students' files, making TrustVault a trusted, transparent and paperless verification system for the department.

VI. Objectives of the Proposed Application

- To develop a secure, college-exclusive document vault for MCA students.
- To protect every document with AES-256-GCM encryption and a vault PIN.
- To allow access only through three-factor sign-in (password, email code, TOTP).
- To verify documents automatically using format rules, checksums and OCR.
- To let faculty request documents from a student or a batch, with consent.
- To allow each shared document to be opened only once, within 24 hours.
- To keep a tamper-evident record of every action in a hash-chained ledger.
- To reduce paperwork and manual verification in the department.
- To build a responsive web application that also installs on Android phones.

VII. Scope / Use

Target Environment

Designed for the Department of Computer Applications (MCA students, faculty and office staff), adaptable to other departments and colleges.

Secure Digital Vault

Keeps identity documents (PAN, Aadhaar, licence) and academic records (marksheets, certificates) in one encrypted place.

Automatic Verification

Checks documents in seconds without a manual verifier and tells students what to fix.

Consent-Based Sharing

Faculty receive documents for placements, scholarships and records only when the student agrees.

Transparency & Privacy

An audit log and a tamper-evident ledger record every action; consent and account deletion follow the DPDP Act 2023, with future expansion to employers.

VIII. Technology to be used (Specific Framework etc.)

- **Frontend:** React 19 with Vite, Tailwind CSS and shadcn/ui (responsive web interface).
- **Backend:** Node.js with Express 5 (REST API).
- **Database:** MongoDB with Mongoose (users, documents, requests, ledger and audit log).
- **Authentication:** Password (bcrypt), email OTP, Google Authenticator (TOTP), JWT.
- **Security:** AES-256-GCM encryption, SHA-256 ledger, Helmet, rate limiting.
- **Document checks:** Tesseract.js OCR, Sharp and pdf-lib for images and PDFs.
- **Deployment:** Render (cloud hosting), GitHub Actions (CI/CD), Capacitor (Android app).

IX. Functional Requirements of the Application

User Registration & Authentication

- Students register with a valid MX roll number; staff join with a code or invitation.
- Sign-in needs password, email code and Google Authenticator.

Document Vault Management

- Students upload, view and delete documents in Identity, Academic and Other.
- Files are encrypted, numbers are masked and a PIN unlocks the vault.

Automatic Document Verification

- Format, checksum, OCR and originality checks give a score out of 100.
- Verified only when every check passes; failed checks are shown.

Faculty Requests & Consent

- Faculty request documents from one student or a whole batch (e.g. 26MX).
- Students share all, some or none; each document opens once in 24 hours.

Administration & Audit

- The administrator manages users, roles and invitations and the audit log.
- A ledger integrity check detects any changed file or record.

Notifications & Export

- Email alerts for new devices, requests and document views.
- Documents are exported as a password-protected ZIP by email.

X. Non-Functional Requirements of the Application

Security

- Encryption at rest (AES-256-GCM) and in transit (HTTPS).
- Rate limiting, account lockout and one-time codes.

Performance

- Pages load on demand; documents are checked in seconds.

Usability

- Responsive interface for desktop and mobile, light and dark themes.

Reliability

- Automated unit, end-to-end and browser tests on every change.

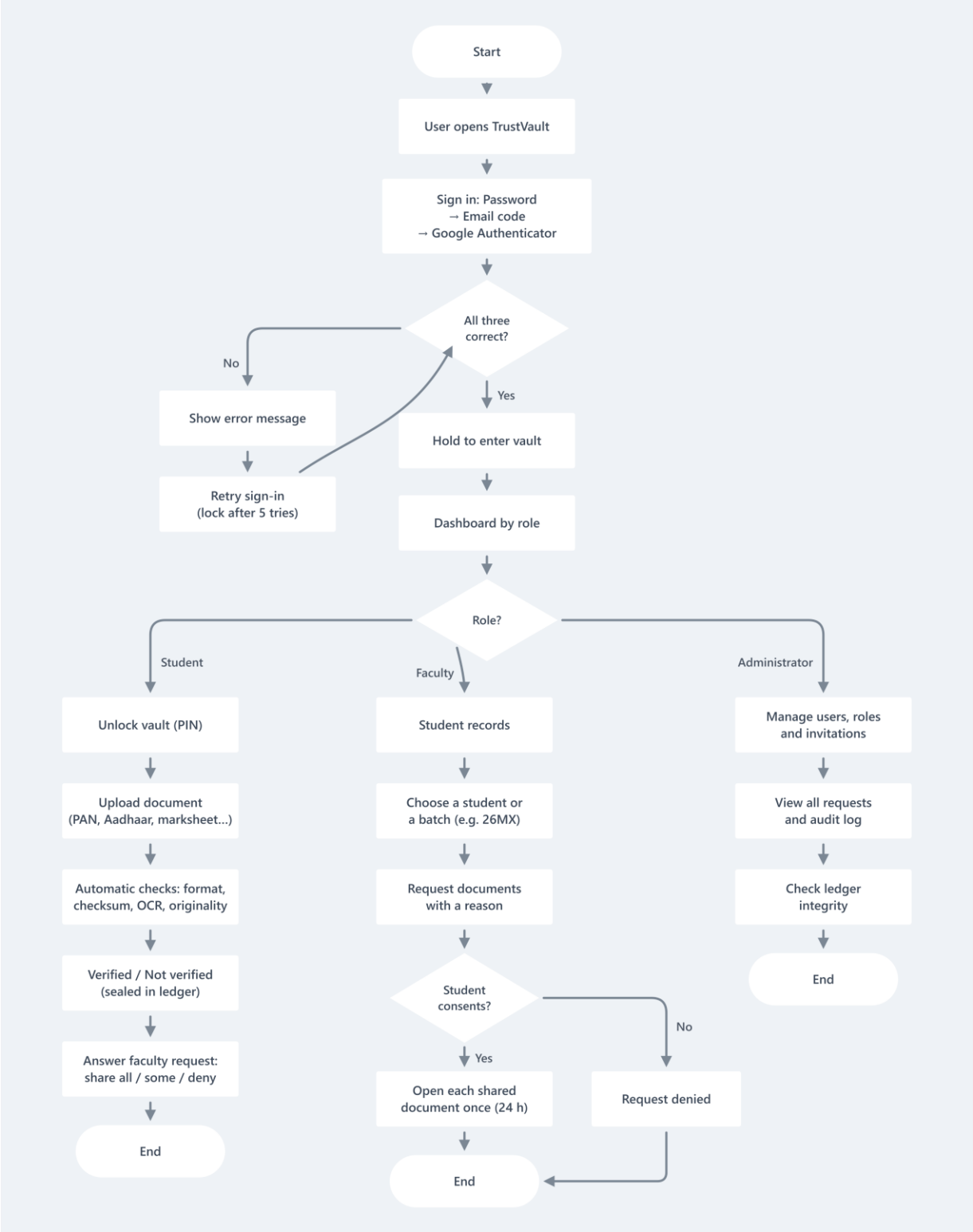
Maintainability

- Modular frontend and backend with a CI/CD pipeline.

Privacy

- Consent before sharing and the right to delete the account (DPDP Act 2023).

XI. Flow diagram of the Application (CAD/ Flow chart kind of simple diagram to illustrate the functional flow of your application)



XII. Database Schema of the application (if any):

Users Collection

Field	Type	Description
_id	ObjectId (PK)	Unique user identifier
name	String	Full name
email	String	Login credential (unique)
studentId	String	MX roll number (students)
passwordHash	String	bcrypt hash of the password
role	String	Student / Faculty / Administrator
totpSecret	String	Encrypted Authenticator secret
vaultPinHash	String	Hash of the vault PIN

Documents Collection

Field	Type	Description
_id	ObjectId (PK)	Unique document ID
owner	ObjectId (FK)	Student reference
type	String	PAN / Aadhaar / Licence / Marksheet / etc.
title	String	Document name
numberMasked	String	Masked number (XXXX XXXX 1234)
storedName	String	Encrypted file on disk
sha256	String	File fingerprint
status	String	Pending / Verified / Not verified / Stored
autoCheck	Object	Score and result of each check
createdAt	Date	Upload time

Access Requests Collection

Field	Type	Description
_id	ObjectId (PK)	Unique request ID
verifier	ObjectId (FK)	Faculty reference
student	ObjectId (FK)	Student reference
reason	String	Purpose of the request
types	Array	Document types asked for
granted	Array	Types the student shared
opened	Array	Documents already opened (once each)
status	String	Pending / Approved / Denied / Revoked / Used
expiresAt	Date	End of the 24-hour access

Ledger (Blocks) Collection

Field	Type	Description
index	Number (PK)	Block number
action	String	Upload / Verified / Not verified
docId	ObjectId (FK)	Document reference
fileHash	String	SHA-256 of the file
prevHash	String	Hash of the previous block
hash	String	Hash of this block
at	Date	Time recorded

Audit Log Collection

Field	Type	Description
_id	ObjectId (PK)	Unique entry ID
user	ObjectId (FK)	Who performed the action
subject	ObjectId (FK)	Whose data it was
action	String	Event name
createdAt	Date	Event time

XIII: Timeline of Activities planned/completed

Phase	Activity	Status
Week 1	**Requirement Analysis**	Completed
Week 2	**UI/UX Design**	Completed
Week 3	**Database Design**	Completed
Week 4	**Frontend Development**	Completed
Week 5	**Backend Development**	Completed
Week 6	**Integration & Testing**	Completed
Week 7	**Bug Fixing & Optimization**	Completed
Week 8	**Documentation & Final Review**	In progress

Signature of the students

Name 1: KIRUBHAKARAN K (26MX325)

Name 2: POOJA M (26MX337)

Name 3: SRINITHISH J K (26MX347)

Signature of the Guide

Name: DR. ILAYARAJA N